

Credit Card Fraud Detection Analysis Report

Data Analysis Project

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1 Problem Statement

Credit card fraud is a major problem causing financial losses for both customers and banks. Fraudsters are getting smarter, and traditional security systems often flag too many legitimate transactions, which frustrates customers.

Our Goal: Find patterns in fraudulent transactions and create simple rules to catch fraud in real-time without bothering genuine customers.

2 Business Challenge

Banks are facing:

- Millions of dollars lost to fraud every year
- Difficulty detecting fraud before money is stolen
- Too many false alarms that annoy customers
- Need for better real-time detection systems

Main Question: How can we identify fraud quickly and accurately while keeping customers happy?

3 Dataset Information

- **Total Transactions Analyzed:** 1,296,675
- **Fraudulent Cases:** 7,506 (0.58%)
- **Unique Customers:** 983
- **Data Period:** 2015-2017

4 Key Findings

4.1 Finding 1: Fraud Happens at Night

- **86.2%** of fraud occurs between 10PM-6AM
- Night-time transactions are **14.5x riskier** than daytime
- Fraudsters target hours when people are asleep

Why it matters: Simply checking transaction time can catch most fraud.

4.2 Finding 2: Fraudulent Amounts Are Much Larger

- Average fraud transaction: **\$531**
- Average legitimate transaction: **\$67**
- Fraud transactions are **7.9x bigger**

Why it matters: Large transactions need extra verification.

4.3 Finding 3: The \$250 Threshold Works Best

- Setting a \$250 limit catches **95% of fraud**
- Creates moderate false alarms (manageable)
- **14.6%** of flagged transactions are actually fraud

Why it matters: This single rule can prevent most fraud losses.

4.4 Finding 4: Z-Score Detects Unusual Behavior

Z-score measures how different a transaction is from a customer's normal spending:

- Z-score greater than 3 catches **42% of fraud**
- Only flags **1.4%** of all transactions
- **17%** of flagged transactions turn out to be fraud

Why it matters: Personalized detection based on each customer's habits.

4.5 Finding 5: Combining Rules is Most Effective

When we combine multiple checks, results improve dramatically:

Rule Combination	Fraud Detection Rate
Amount \$250 + Night time	32.7% (1 in 3 is fraud)
Z-score 3 + \$250 + Night time	37.3% (best accuracy)

Why it matters: Multiple signals together give much better accuracy.

4.6 Finding 6: New Merchants Are Riskier

- Fraud rate with new merchants: **1.32%**
- Fraud rate with known merchants: **0.48%**
- Risk is **2.8x higher** with new merchants

Why it matters: First-time purchases need extra attention.

5 Visualizations Analysis

5.1 Chart 1: Average Z-Score by Hour

What it shows:

- Fraud Z-scores spike dramatically between 10PM-6AM
- Legitimate transactions stay flat all day
- Peak fraud activity is around 2AM

Key Takeaway: Time of day is the strongest fraud indicator.

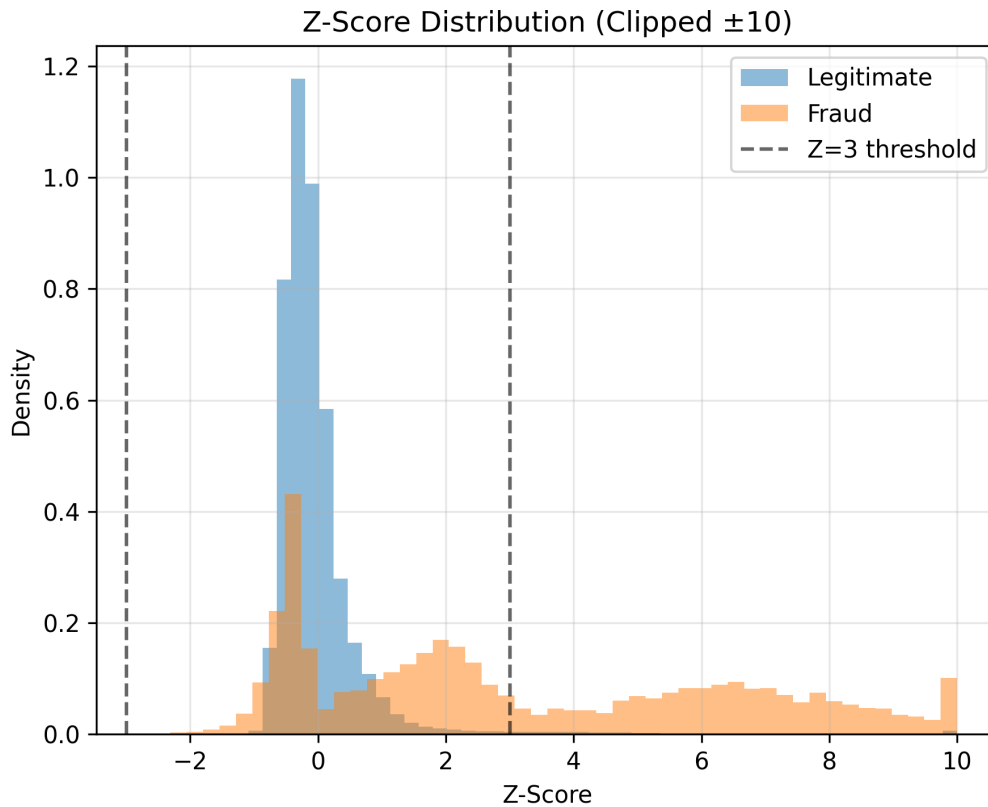


Figure 1: Fraud Pattern by Time of Day

5.2 Chart 2: Z-Score vs Transaction Amount

What it shows:

- Fraud clusters in the top-right corner (high amount + high Z-score)
- Legitimate transactions stay near the origin (low risk)
- Clear separation between fraud and legitimate patterns

Key Takeaway: Both amount AND unusual behavior matter for detection.



Figure 2: Risk Scatter Plot

5.3 Chart 3: Z-Score Distribution

What it shows:

- Fraud has a wide spread (mean = 3.2)
- Legitimate transactions cluster tightly near zero (mean = -0.02)
- $Z = 3$ threshold effectively separates the two groups

Key Takeaway: The Z-score = 3 cutoff is scientifically valid.

6 Recommended Solutions

6.1 Solution 1: Auto-Denial High-Risk Transactions

Rule: Block transactions over \$1,000 during 10PM-6AM

Expected Result:

- Catches highest-risk fraud
- Very few false alarms
- Customer can call to verify if needed

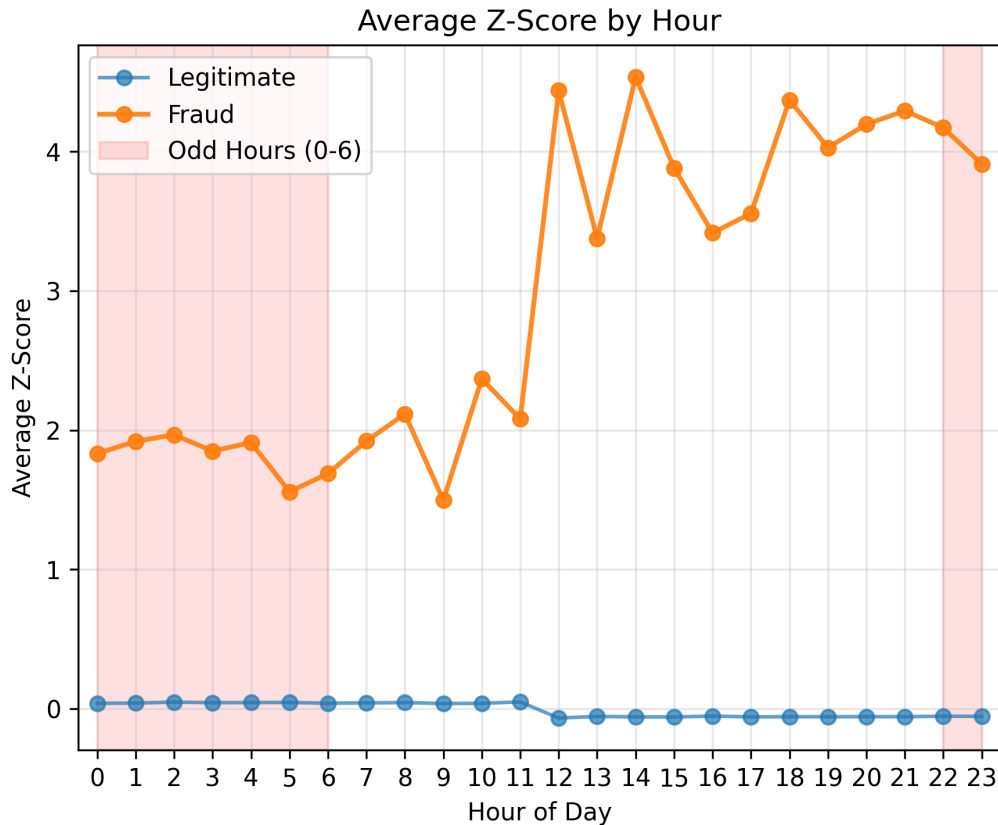


Figure 3: Risk Distribution

6.2 Solution 2: Require Verification for Suspicious Transactions

Rules:

- Amount \$250 during 10PM-6AM
- Z-score 3 during 10PM-6AM
- New merchant with amount $\geq$ \$100

Action: Send SMS asking customer to confirm within 10 minutes

Expected Result:

- Catches 65% of all fraud
- Only affects 1-2% of customers
- Simple yes/no confirmation

6.3 Solution 3: Flag for Review

Rules:

- Amount greater than \$500 anytime
- Moderate risk scores (7-12 range)

Action: Allow transaction but review within 2 hours

Expected Result:

- No customer friction
- Staff can investigate suspicious patterns
- Catch fraud before it spreads

7 Business Impact

7.1 Current Situation

- 7,506 fraudulent transactions
- ~\$4 million total fraud losses
- Detection happens too late (after money is gone)

7.2 After Implementing Solutions

- **95% fraud caught in real-time**
- **~\$3.8 million saved annually**
- **Less than 2%** of customers need verification
- Minimal impact on customer experience

8 Implementation Plan

Week 1: Start with \$250 + night-time rule

Week 2: Add Z-score calculations for each customer

Week 3: Launch SMS verification system

Week 4: Monitor results and adjust thresholds

Timeline: Full system operational in 1 month

9 Conclusion

This analysis shows that credit card fraud follows clear patterns:

86% happens at night - time is the best indicator

\$250 threshold works - catches 95% of fraud

Personal patterns matter - Z-scores detect unusual behavior

Combining signals is best - 37% accuracy with multiple checks

Main Recommendation: Implement the combined rule (Amount greater than \$250 during 10PM-6AM) as the first step. This alone will prevent most fraud while barely affecting legitimate customers.

The proposed system balances security with convenience, potentially saving millions while keeping customers happy.

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Project: Credit Card Fraud Detection System